Hanzo Platform Integration

with Digital Securities

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Abstract

We describe how the Hanzo platform infrastructure—Base, IAM, KMS, Gateway—powers the ArcaLabs digital securities stack. Digital securities require investor onboarding with KYC/AML verification, trade execution through a regulated alternative trading system (ATS), institutional MPC custody, and comprehensive reporting. Rather than building these capabilities from scratch, ArcaLabs integrates Hanzo's existing platform services to provide identity management, secret management, API routing, and data persistence. We detail the full architecture: investor onboarding via Simplici KYC integrated through Hanzo IAM, trade execution through the the regulated ATS with Hanzo Gateway routing, MPC custody with keys managed by Hanzo KMS, and reporting pipelines built on Hanzo Base. This integration demonstrates the composability of the Hanzo platform for regulated financial applications.

1 Introduction

The issuance and trading of digital securities requires infrastructure that spans identity management, cryptographic key management, API gateway services, data persistence, and regulatory reporting. Building each component from scratch would duplicate capabilities that already exist in production within the Hanzo platform.

Hanzo has operated as an AI-powered commerce and infrastructure platform since its founding (Techstars '17), serving thousands of applications through its unified service stack. The platform provides five core services relevant to digital securities:

1. **Hanzo IAM**: Identity and access management with OIDC, OAuth 2.0, and org-scoped permissions.
2. **Hanzo KMS**: Secrets and key management with envelope encryption.
3. **Hanzo Base**: Application runtime with embedded SQLite (local dev) or PostgreSQL (production).
4. **Hanzo Gateway**: API gateway with rate limiting, request routing, and TLS termination.
5. **Hanzo Webhooks**: Event delivery with guaranteed at-least-once semantics.

ArcaLabs consumes these services to build its digital securities platform without reimplementing foundational infrastructure.

2 Architecture Overview

3 Investor Onboarding via Hanzo IAM

Investor onboarding combines Hanzo IAM identity management with Simplici KYC/AML verification.

3.1 Identity Creation

1. Investor registers via ArcaLabs platform. Registration request routed through Hanzo Gateway.
2. Hanzo IAM creates an identity record with OIDC subject identifier. The investor receives OAuth 2.0 credentials.

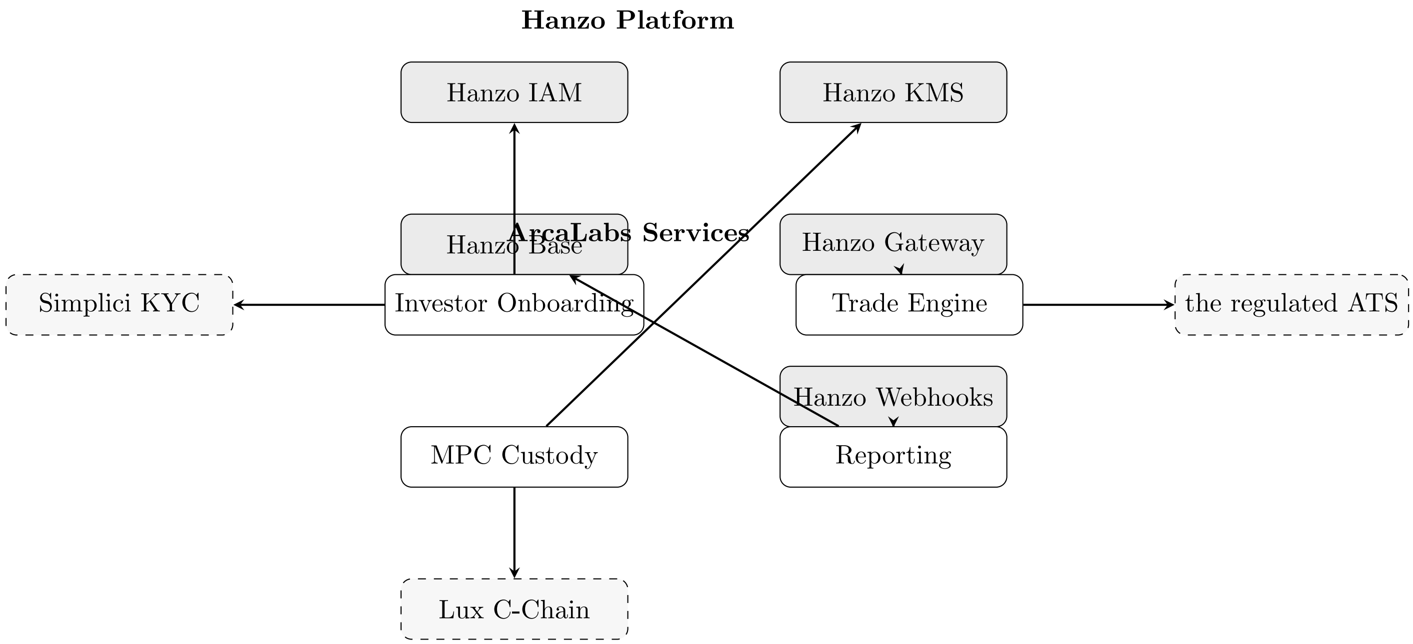

Figure 1: ArcaLabs digital securities architecture on Hanzo platform

3. IAM assigns the investor to the ArcaLabs organization with the `investor` role.
4. All subsequent API calls include the IAM bearer token. Gateway extracts the org from the JWT `owner` claim and scopes all data queries to that org.

3.2 KYC/AML via Simplici

1. ArcaLabs onboarding service redirects the investor to Simplici's verification flow.
2. Simplici performs: identity document verification, liveness detection, OFAC/sanctions screening, PEP screening, and accredited investor verification.
3. Upon completion, Simplici returns a signed attestation to ArcaLabs via webhook.
4. ArcaLabs compliance service validates the attestation and updates the investor's IAM profile with: KYC status, accreditation status, accreditation expiry, jurisdiction.
5. The compliance service writes the investor to the on-chain `ComplianceRegistry` whitelist on Lux C-Chain.

3.3 Ongoing Monitoring

Simplici performs continuous monitoring:

- Daily sanctions list re-screening.
- Adverse media monitoring.
- Annual accreditation re-verification trigger.
- Alerts delivered to ArcaLabs via Hanzo Webhooks.

4 Trade Execution via the regulated ATS

Secondary trading of digital securities occurs through the regulated platform ATS (Alternative Trading System), a SEC-registered trading venue.

4.1 Order Flow

1. Investor submits order via ArcaLabs trading interface.
2. Request authenticated by Hanzo IAM, routed through Hanzo Gateway to the ArcaLabs trade engine.
3. Trade engine validates: investor whitelist status, lockup compliance, accreditation validity.
4. Approved order forwarded to the regulated ATS matching engine.
5. ATS matches buy and sell orders. Matched trades return to ArcaLabs for settlement.
6. Settlement: atomic swap of security tokens and USDC on Lux C-Chain via MPC-signed transaction.

4.2 Gateway Configuration

Hanzo Gateway handles the routing between ArcaLabs services and the the regulated ATS:

- **Rate limiting**: Per-investor and per-org request limits to prevent abuse.
- **Authentication**: IAM token validation at the gateway layer. Invalid tokens rejected before reaching backend services.
- **TLS**: End-to-end encryption. Mutual TLS between Gateway and ATS.
- **Audit logging**: Every request logged with investor identity, timestamp, and response code.

5 MPC Custody via Hanzo KMS

MPC key material is managed through Hanzo KMS (kms.hanzo.ai):

5.1 Key Hierarchy

1. **KMS Master Key**: Root encryption key managed by Hanzo KMS. Never exported.
2. **MPC Share Encryption Keys**: Per-share-holder encryption keys derived from the KMS master key via envelope encryption.
3. **MPC Key Shares**: The actual CGGMP21 key shares, encrypted at rest with their respective share encryption keys.

5.2 Operational Flow

1. **DKG**: During distributed key generation, each MPC node requests a share encryption key from KMS. The generated key share is immediately encrypted and stored.
2. **Signing**: When a signing session begins, the MPC node requests decryption of its key share from KMS. KMS validates the request against IAM identity and returns the decrypted share. The share exists in memory only during the signing session.
3. **Rotation**: During key resharing, new shares are encrypted with fresh KMS-derived keys. Old encrypted shares are securely deleted.
4. **Audit**: Every KMS access is logged: who requested what key, when, and for what purpose.

6 Data Persistence via Hanzo Base

Hanzo Base provides the data layer for ArcaLabs services:

Investor records KYC status, accreditation, jurisdiction, wallet addresses. Linked to IAM identity.

Trade records Order submissions, matches, settlements. Immutable audit trail.

Compliance events Whitelist changes, restriction violations, corporate actions. Indexed for regulatory reporting.

NAV history Daily NAV calculations for ArCoin and other fund products.

MPC session logs Signing requests, policy decisions, completion status. Required for SEC 17a-4 compliance.

Base configuration:

- **Local development**: SQLite embedded in Base runtime. Zero external dependencies.
- **Production**: PostgreSQL backend with Base runtime providing the application layer, authentication (via IAM), and real-time subscriptions.
- **Replication**: PostgreSQL streaming replication for high availability.

7 Reporting Pipeline

ArcaLabs' reporting requirements span regulatory filings, investor communications, and internal audit:

7.1 Regulatory Reports

- **Form D filings**: Generated from issuance records in Base. Auto-populated with investor counts, amounts raised, exemption type.
- **Blue sky filings**: State-level notice filings based on investor jurisdiction data.
- **ATS Form ATS-N**: Quarterly filing for the regulated platform ATS, generated from trade records.
- **FINRA trade reports**: Real-time trade reporting generated from settlement events.

7.2 Investor Communications

- **Account statements**: Monthly statements generated from Base records, delivered via Hanzo Webhooks to investor email.
- **Tax documents**: K-1 (fund interests) and 1099 (interest/dividends) generated annually.
- **NAV reports**: Daily NAV publications for fund products.

7.3 Event Pipeline

1. On-chain events (transfers, compliance decisions, corporate actions) are indexed by an event listener.
2. Events written to Base with chain metadata (block number, tx hash, timestamp).
3. Hanzo Webhooks delivers events to configured endpoints (reporting systems, compliance dashboards, investor portals).
4. Events are immutable in Base—no deletion, only append.

8 Full Architecture Diagram

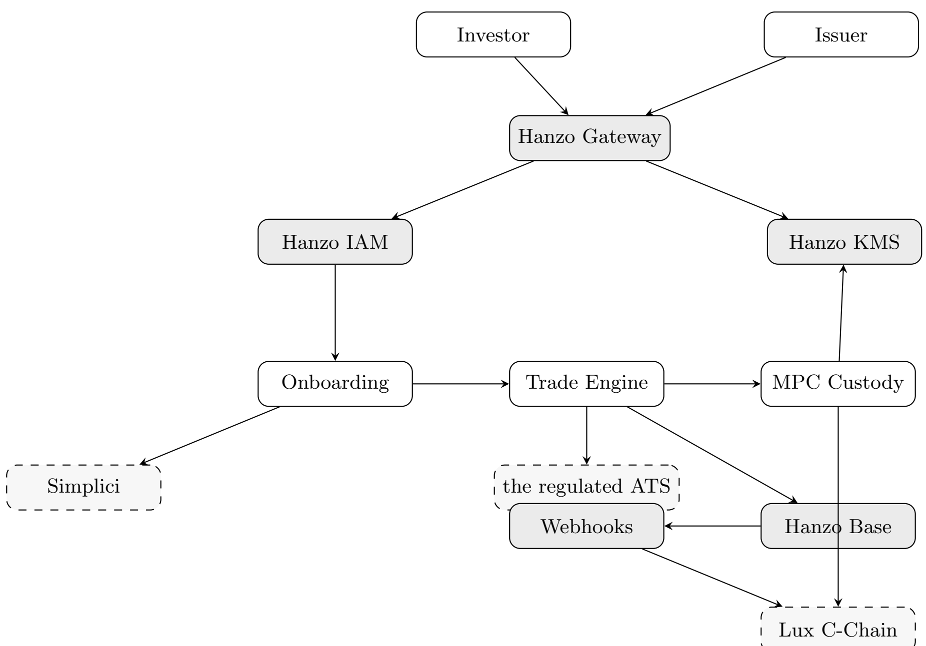

Figure 2: Complete system architecture: Hanzo platform + ArcaLabs services + external dependencies

9 Conclusion

The Hanzo platform provides the foundational infrastructure for ArcaLabs' digital securities operations. IAM handles investor identity and access control with org-scoped permissions. KMS secures

MPC key material with envelope encryption and comprehensive audit logging. Base provides the data persistence layer with production-grade PostgreSQL and development-friendly SQLite. Gateway routes and rate-limits all traffic with IAM token validation. Webhooks delivers lifecycle events to reporting and compliance systems.

This integration demonstrates that regulated financial applications can be built by composing existing platform services rather than reimplementing infrastructure. The Hanzo platform's service boundaries align with the separation of concerns required by securities regulation: identity (IAM), custody (KMS + MPC), data (Base), and communication (Gateway + Webhooks).

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